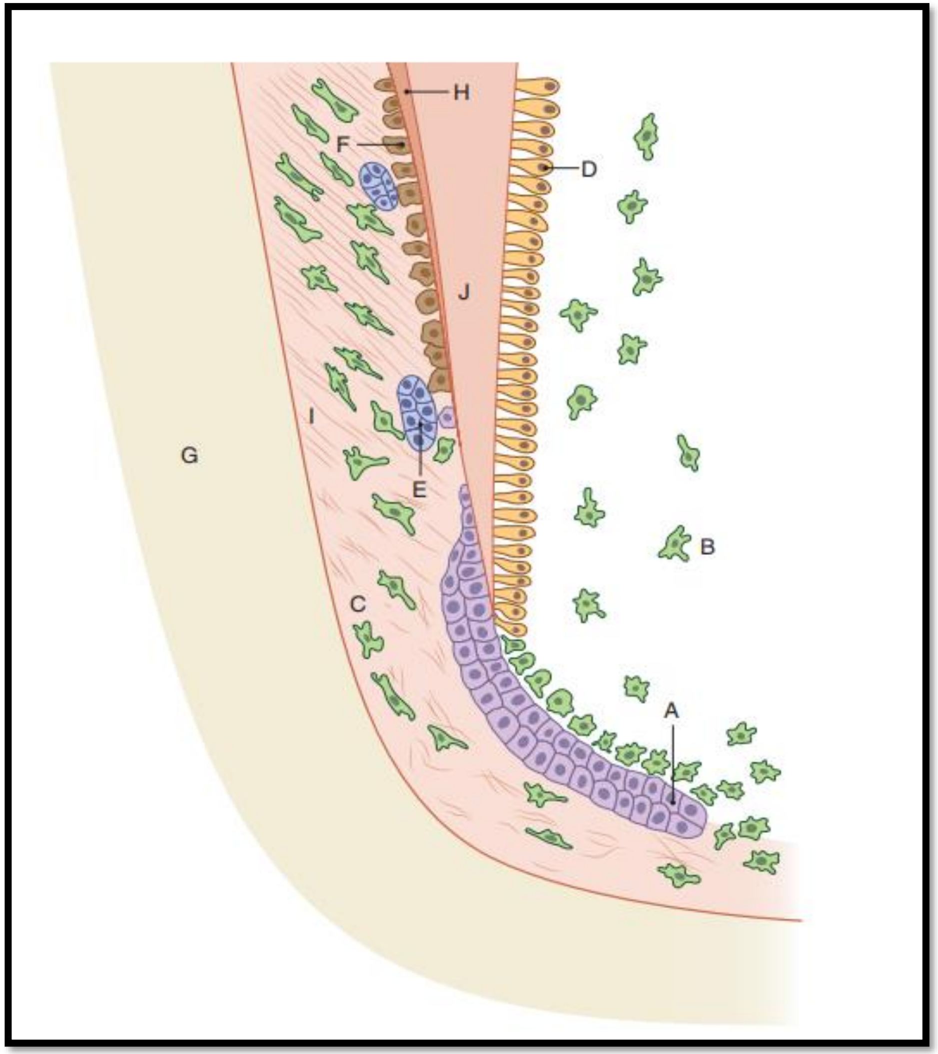
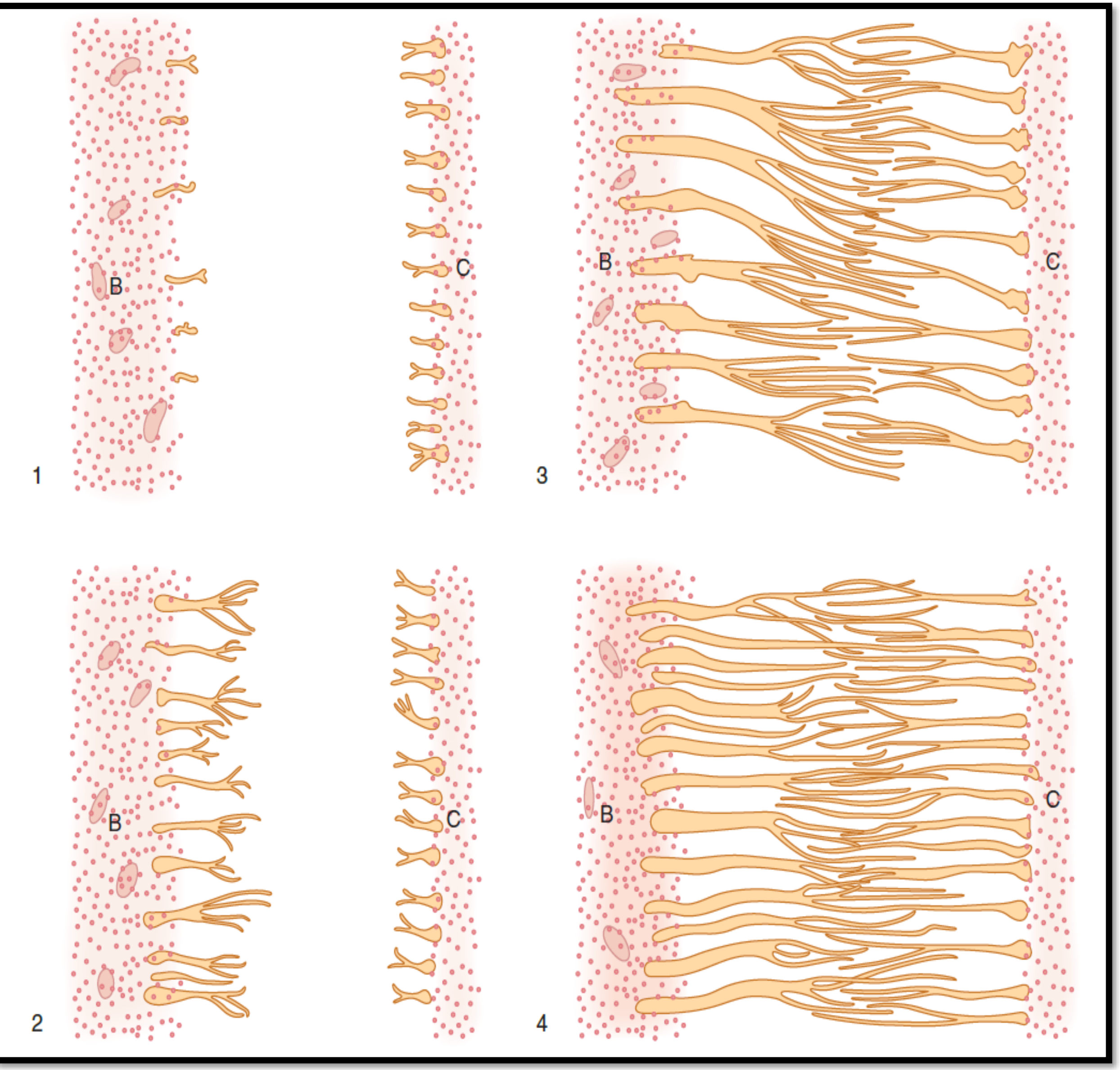
DEVELOPMENT OF ROOT & PDL

TASK: PASTE, IDENTIFY AND LABEL THE FOLLOWING FIGURES



Schematic drawing of the developing root.

A = epithelial root sheath;
B = dental papilla;
C = dental follicle;
D = odontoblasts;
E = epithelial rests;
F = cementoblasts;
G = developing alveolar bone;
H = developing cementum;
I = developing periodontal ligament;
J = root dentine.



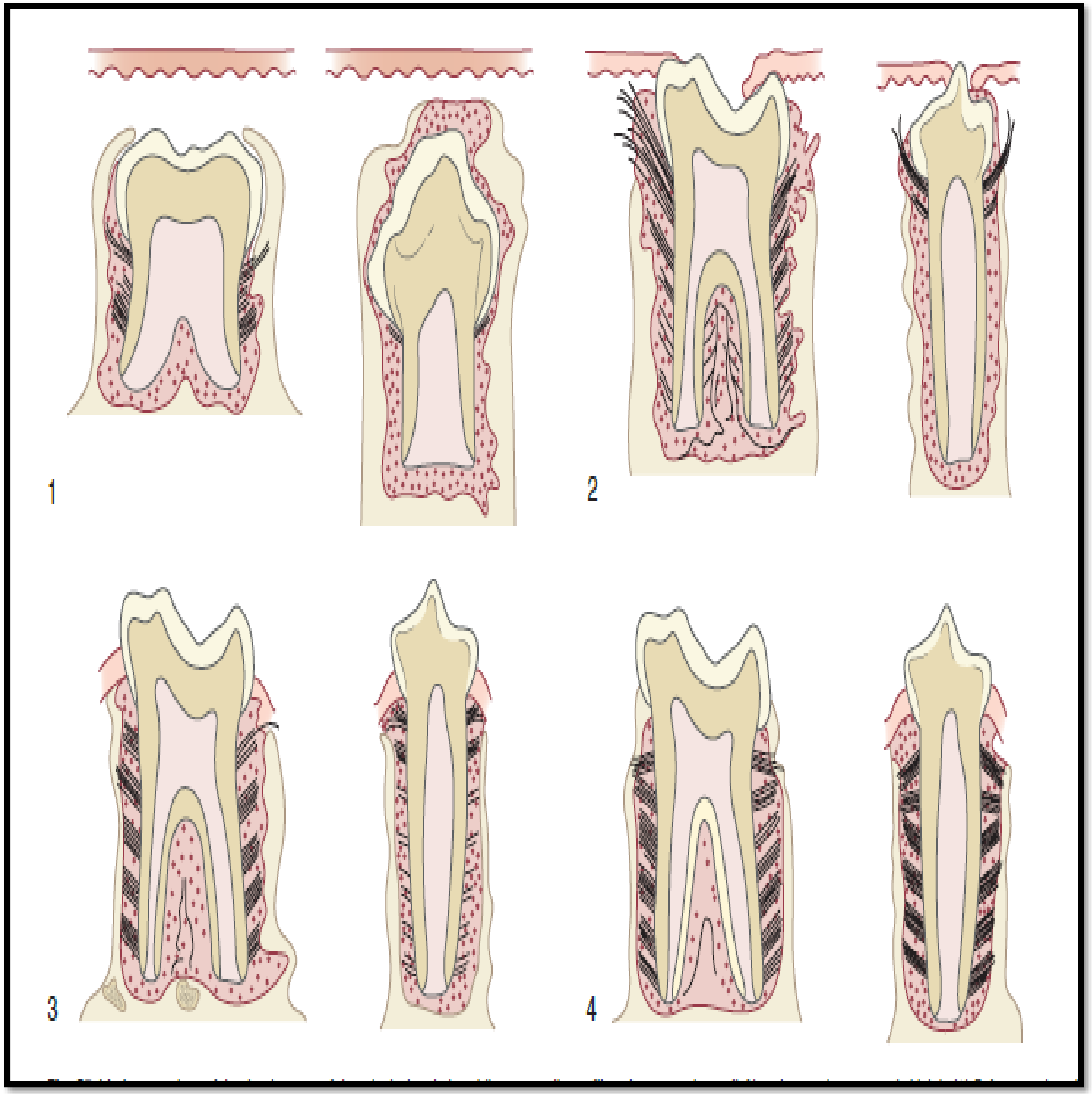
Summary of the development of collagen fibres of the periodontal ligament.

(1) Fine brushlike fibres are first seen emanating from cementum (C). Only a few fibres project from the alveolar bone (B) and extend into the unorganised collagenous elements that occupy the broad central zone of the developing periodontal ligament.

(2) Sharpey's fibres, thicker and more widely spaced than those of cementum, emerge from the bone to extend towards the tooth and appear to unravel as they arborise at their ends. The closely spaced, cemental fibres are still short, giving the root a brushlike appearance.

(3) The alveolar fibres extend further into the central zone to join the lengthening cemental fibres.

(1) With occlusal function, the principal fibres become classically organised, thicker and continuous between bone and cementum



A comparison of the development of principal PDL collagen fibres b/w primary (left) & succedaneous teeth (right).

(1) Before eruption

The dentogingival & oblique PDL fibres are well developed in permanent molar. In the permanent premolar, however, only the dentogingival fibres are organised, the developing PDL being composed of loosely structured collagenous elements.

(2) As the tooth emerges into the oral cavity,
The PDL of the permanent molar is well differentiated, the oblique fibres being the most conspicuous. However, at this stage in the permanent premolar only the fibres in the region of the alveolar crest are becoming organised. In the PDL itself, although collagen fibres are developing, they do not yet span the PDL space.

(3) On reaching occlusion, the fibre groupings in the cervical region of the permanent molar now become organised. In the permanent premolar, while the fibre groups cervically appear prominent, those in the apical part of the root appear relatively undeveloped.

(4)After a period in function, the fibres of both the permanent molar and premolar show the classical organisation of the principal fibres

REFERENCE MATERIAL

ROOT FORMATION IMPORTANT POINTS

Root development proceeds → after crown formation & involves:

1) the dental follicle

2) a structure derived from the cervical loop region of the enamel organ called the epithelial root sheath (of Hertwig)

3) the dental papilla.

- The onset of root development coincides with axial phase of tooth eruption.

- $1/4^{th}$ to ½ root present when permanent tooth erupts

- It takes around 3 more years for root completion in a permanent tooth to occur & about 1½ years for root completion in a deciduous tooth.

- **Root extension rates:** 4 & 6 μm/day, but with a rise in rates to a brief peak of 6–10 μm/day about the middle of the root length.

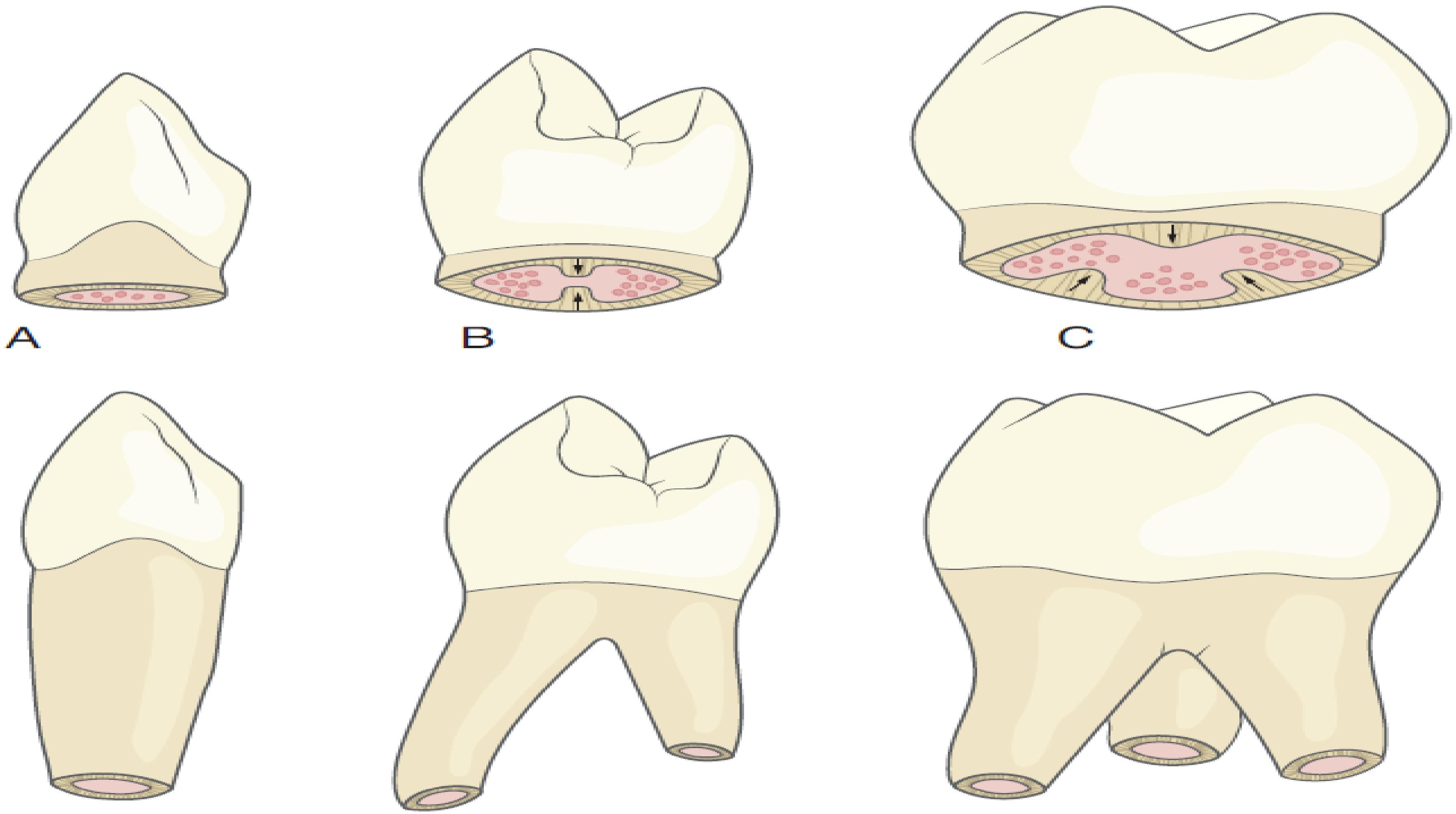


Fig. 25.1 The formation of a single-rooted tooth (A), a two-rooted tooth (B) and a three-rooted tooth (C). Small red circles indicate vascular concentrations.

Cushing Hammock Ligamnet & Pulp Limiting membrane:

Connective tissue lying immediately below the developing root apex was described as fibrous network with fluid filled compartments , connected to the alveolus on either side

It Was thought to provide a strong base to prevent bone resorption by heavy forces of eruption underneath the developing root, this view is no longer held.

It was proven that this fibrous membrane is not attached to bone but merges with fibers of PL , so more correctly was called the **pulp limiting membrane**
Its surgical removal does not affect tooth eruption.

Fig. 25.9 The pulp-limiting membrane (arrowed) (Mallory's trichrome; ×30). Courtesy Dr D Adams.

Table 25.1 Possible phenotypical differences between cementoblasts associated with acellular and cellular cementum

Cementoblasts from acellular cementum	Cementoblasts from cellular cementum
Identifiable for only a short time	Identifiable for longer period
Fibroblast-like morphology	Osteoblast-like morphology
Derived from epithelial root sheath	Derived from mesenchyme
Express cytokeratin	Do not express cytokeratin
Do not express osteocalcin	Express osteocalcin
Do not express receptors for parathormone	Express receptors for parathormone
Do not express TGFβ and IGF	Express TGFβ and IGF
Do not react with E11 antibody	React with E11 antibody

E11 antibody = a marker reacting with osteoblasts and young osteocytes; IGF = insulin-like growth factor; TGF = transforming growth factor.

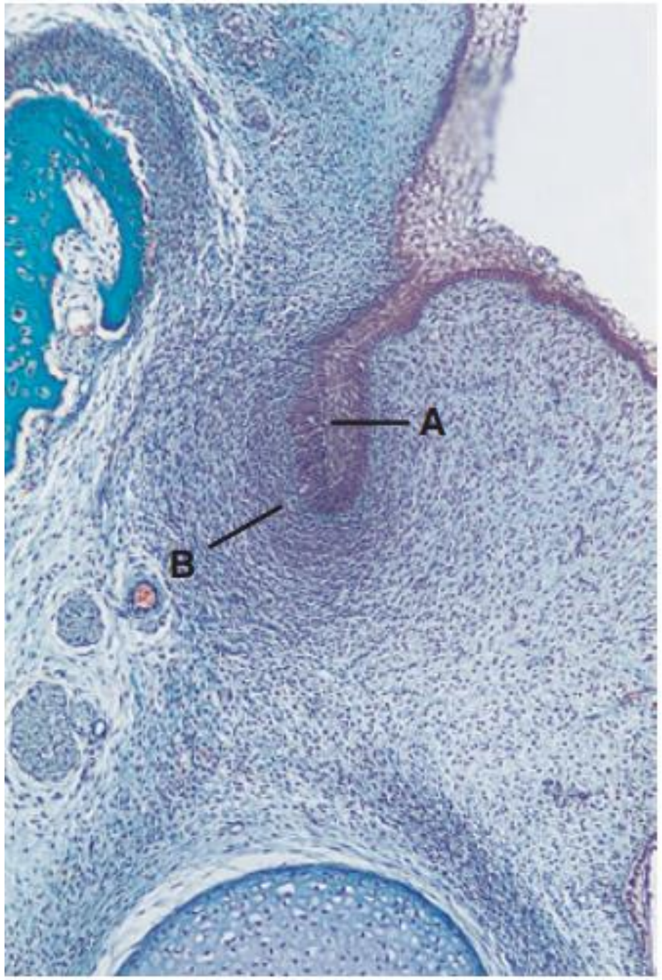


Fig. 24.2 An enamel organ at the cap stage of development (A) surrounding the mesenchymal cells of the dental papilla (B). The tooth germ is surrounded by the developing dental follicle (C) (H & E; ×80). Courtesy Prof. M M Smith.

Fig. 24.1 A developing enamel organ at the early bud stage (A), with a condensation of mesenchymal cells (B) around it (Masson's trichrome stain; ×80).

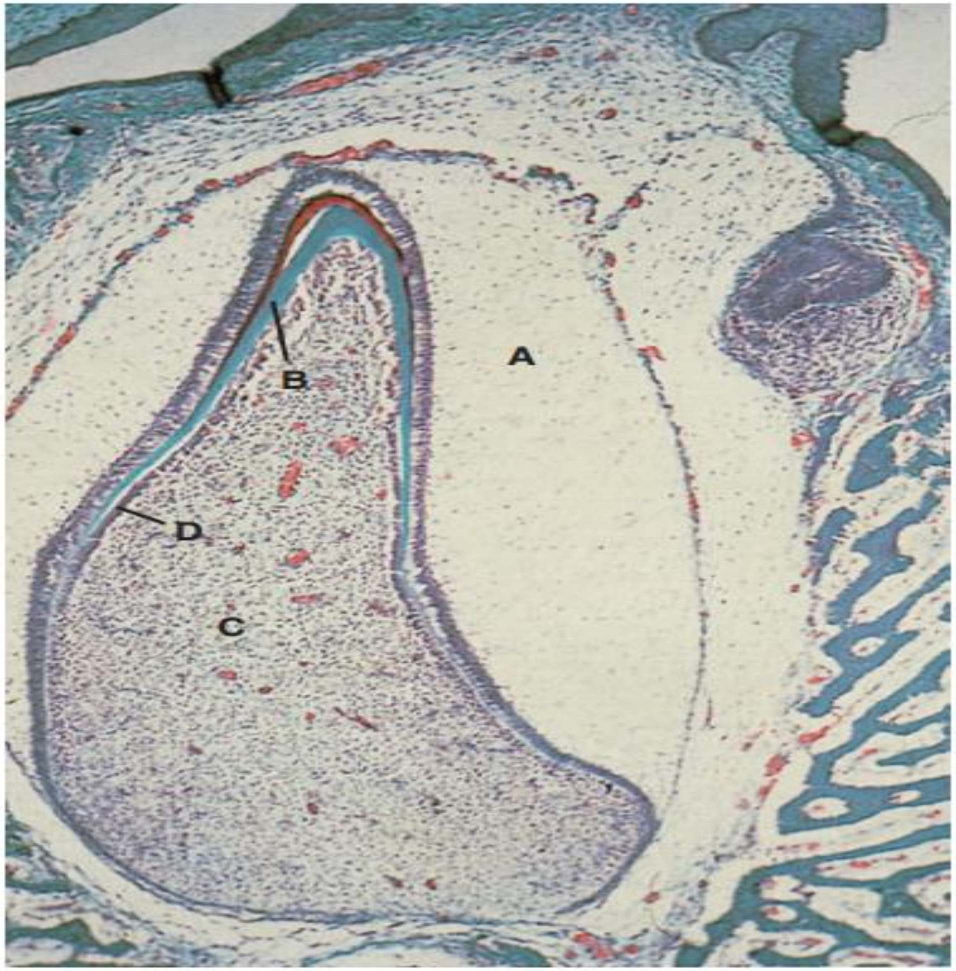


Fig. 24.3 A developing tooth at the late bell stage of development (appositional phase). A = enamel organ; B = developing dentine; C = dental papilla; D = odontoblast layer (Masson's trichrome; ×55).

Development Of The Dental Pulp

- Starts at the **bud stage** as condensation of mesenchymal cells
- Some of these cells will become odontoblasts , those come from the neural crest (ectomesenchyme)
- The others are of mesenchymal origin
- All cells look the same under the microscope
- -These cells rapidly divide and are separated by little extracellular fluid.

Development Of The Dental Pulp

- In the cap stage the cells forming this mass are known as the **dental papilla**
- **Dental follicle** : mesenchymal cells surrounding the developing enamel organ and later on they will form the periodontal ligament and the supporting tooth structures.
- During the bell stage cyto-diffrentiation of cells by signals from internal enamel epithelium take place into :
- 1. Central mass of fibroblasts
- 2. Peripheral layer of odontoblasts.